EART11300

Igneous, sedimentary and metamorphic rocks in hand specimen

Video 1: attributes of **igneous** rocks

Video 2: attributes of **sedimentary** rocks

Video 3: attributes of **metamorphic** rocks

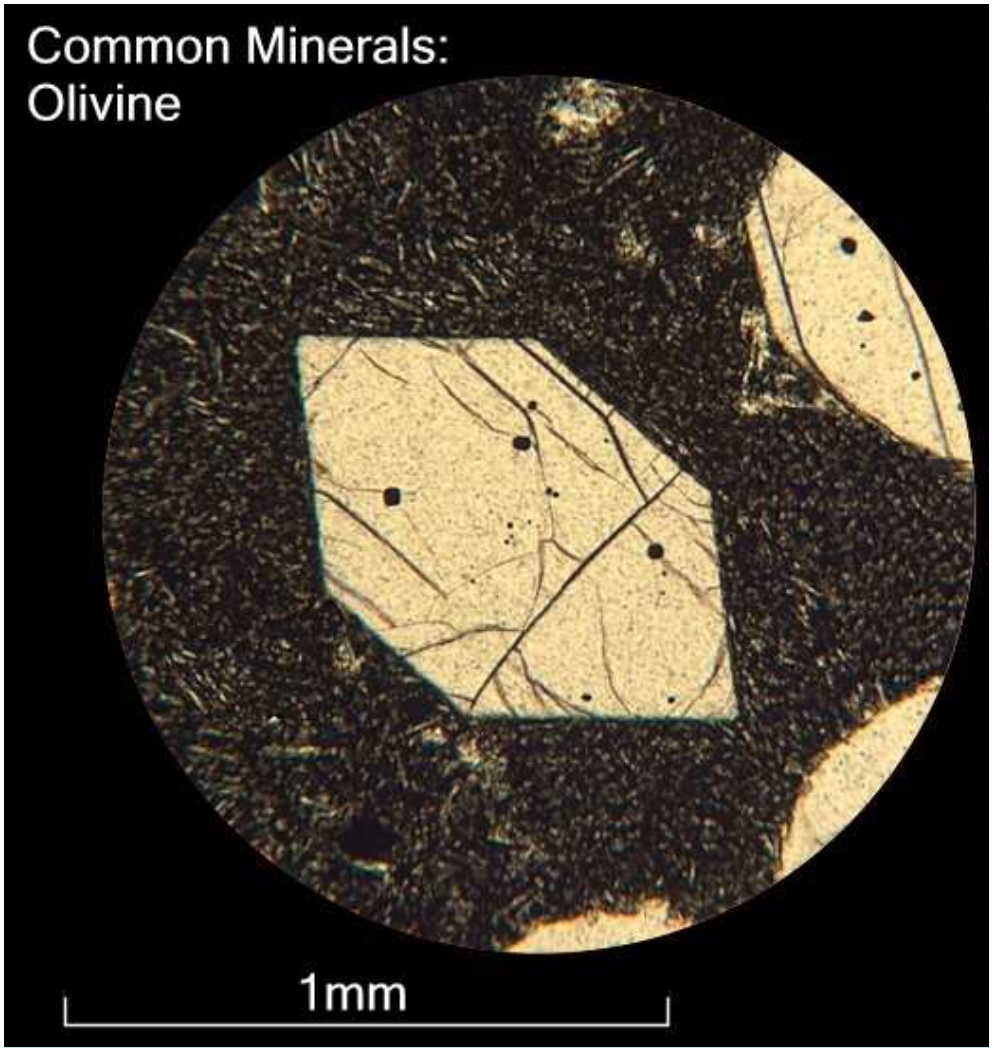
Igneous rocks

When you are looking at a hand specimen or thin section:

- Some crystals have their own shape (i.e. are bounded by flat crystal faces) indicating unhindered growth from a liquid.
- Igneous rocks tend to have interlocking crystals
- Contain relatively few minerals (normally 5 different mineral species or less)
- Crystals of each mineral tend to be quite uniform.
- May contain fossil gas bubbles (= vesicles). Not otherwise porous.
- Crystals don't usually have a preferred orientation (but there are exceptions, e.g. cumulates and rocks with flow-aligned crystals)
- Do not contain organic matter or fossils.

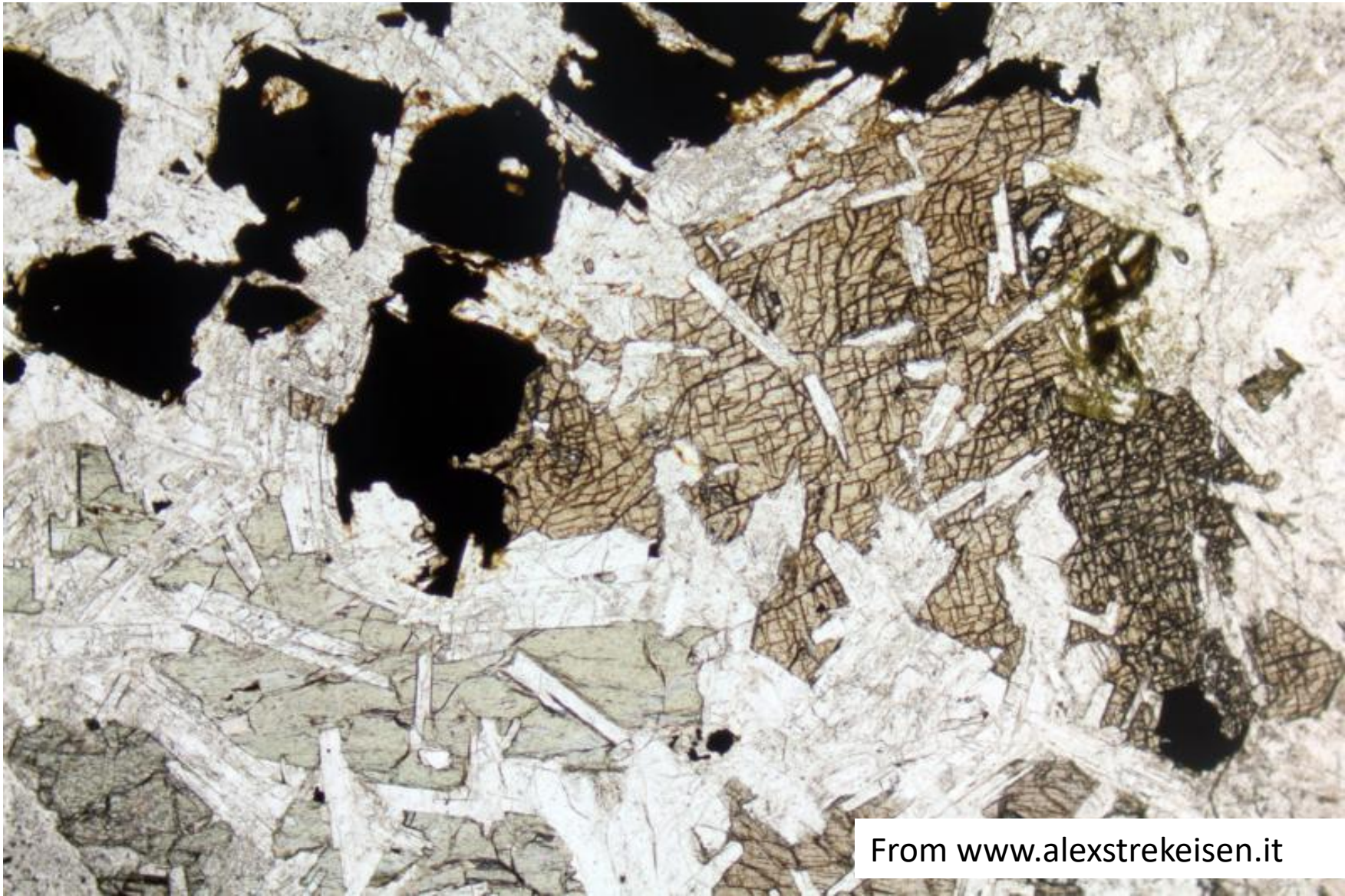
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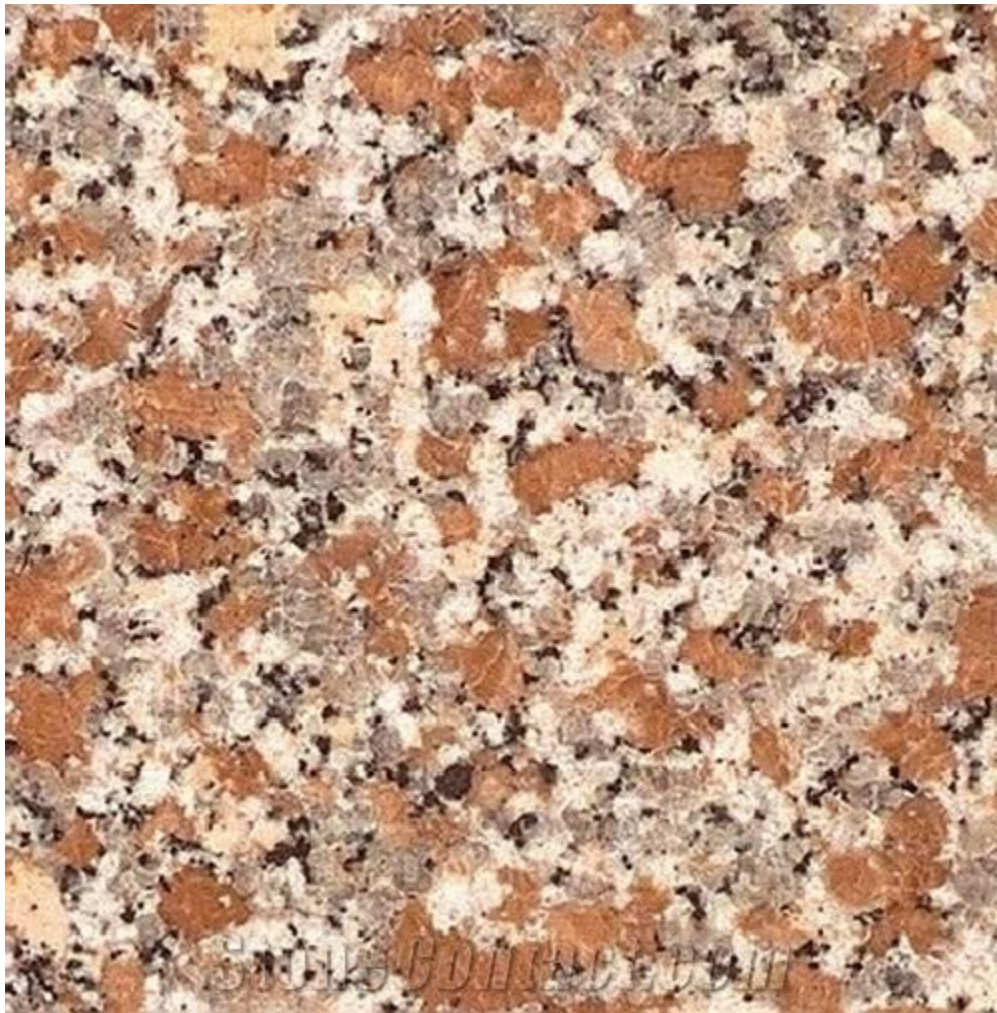
Igneous rocks

- Igneous rocks tend to have interlocking crystals



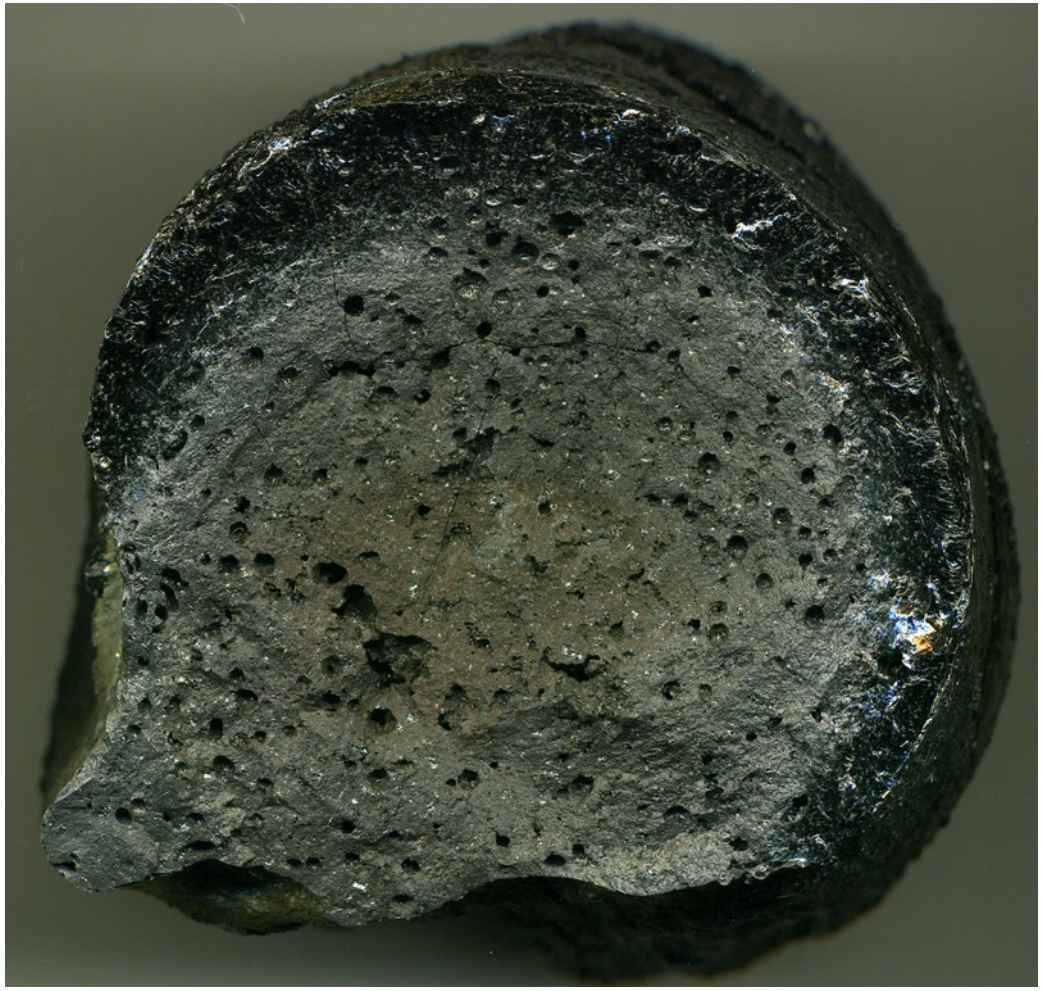
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Igneous rocks

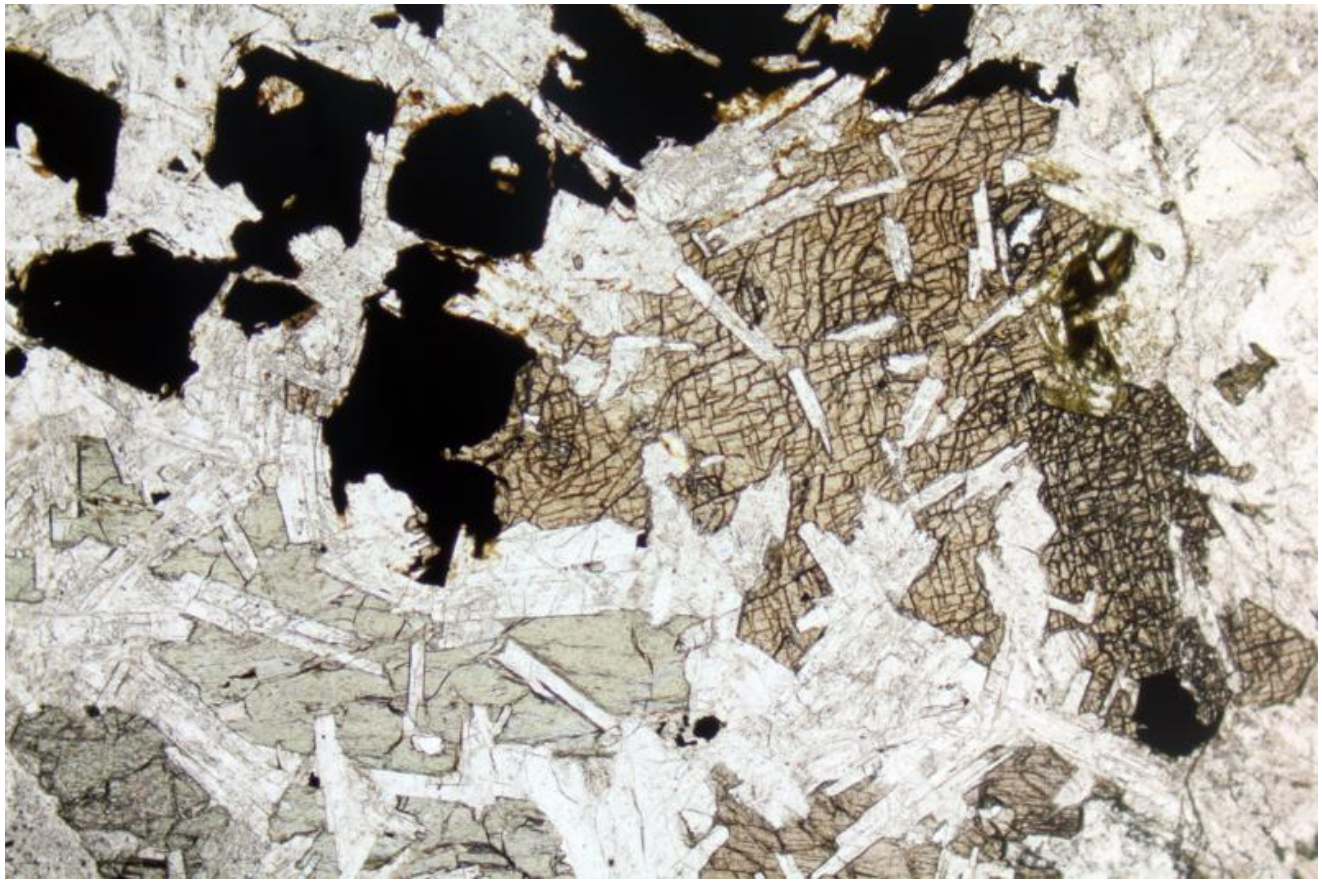
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From Wikipedia

Igneous rocks

- Crystals don’t usually have a preferred orientation (but there are exceptions, e.g. cumulates and rocks with flow-aligned crystals)



Igneous rocks

- Do not contain organic matter or fossils.

Lava flow on Hawaii



From Wikipedia

Weathering of igneous rocks



From Wikipedia



Further classification of the igneous rocks

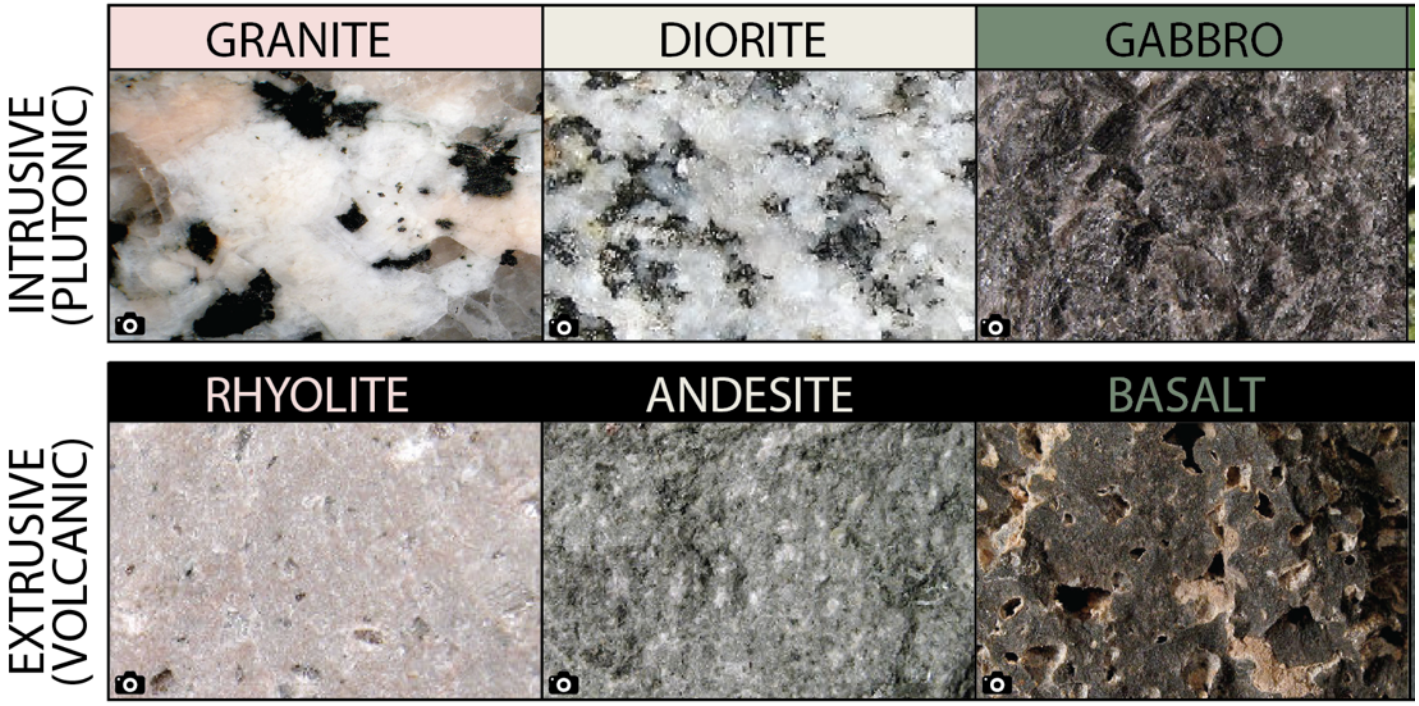
Igneous rocks can be classified in two main ways:

1. Based upon the chemistry and therefore reflected in the mineralogy.
   - High silica = **felsic**. Contains quartz, abundant feldspar, not many mafic (dark coloured) minerals.
   - Intermediate silica = **intermediate**. Contains abundant feldspar, no obvious quartz, some mafic minerals.
   - Low silica = **mafic**. Contains abundant mafic minerals and some feldspar.
   - Very low silica = **ultramafic.** Contains >90% mafic minerals.

Further classification of the igneous rocks

Igneous rocks can be classified in two main ways:

2. Based upon the grain size, which tells us how quickly the rock has cooled:
   - Medium- to coarse-grained = slow cooling = **plutonic**
   - Fine-grained = fast cooling = **volcanic**



https://openpress.usask.ca/app/uploads/sites/29/2018/06/igneous-rock-classification_revised.png